

# Selfhosting / Homeserver

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**SENECA**

Software Engineering Erlangen Camp



# Motivation

- Privacy / Enshittification
- Learning: networks, operations, IaC, etc.
- Saving money?





# Disclaimer

- I am new to selfhosting
- This is my learning experience



# Cloud or Homeserver?



## Homeserver

### Pros

- full control
- privacy: all data is yours
- no vendor lock-in
- low running costs

### Cons

- initial hardware investment costs
- maintenance of all infrastructure
- availability
- scaling



## Cloud

### Pros

- PaaS / SaaS solutions
- availability
- scaling
- no hardware investment

### Cons

- vendor lock-in
- running costs are difficult to predict
- trust issues

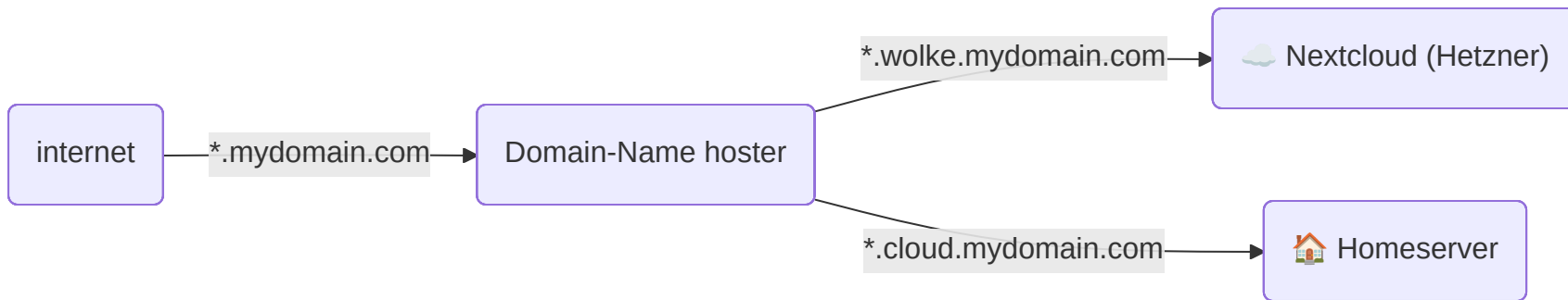


# Typical Applications

| app                 | "before"                 | "after"                |
|---------------------|--------------------------|------------------------|
| calendar            | Google Calendar          | ☁️ Nextcloud (Hetzner) |
| file sharing        | Google Drive             | ☁️ Nextcloud (Hetzner) |
| photos              | Google Photos            | 🏠 <u>immich</u>        |
| document management | private git              | 🏠 <u>Paperless-NGX</u> |
| read-it-later       | Pocket (Mozilla)         | 🏠 <u>Readeck</u>       |
| cooking recipes     | bookmarks, Chefkoch, etc | 🏠 <u>Mealie</u>        |



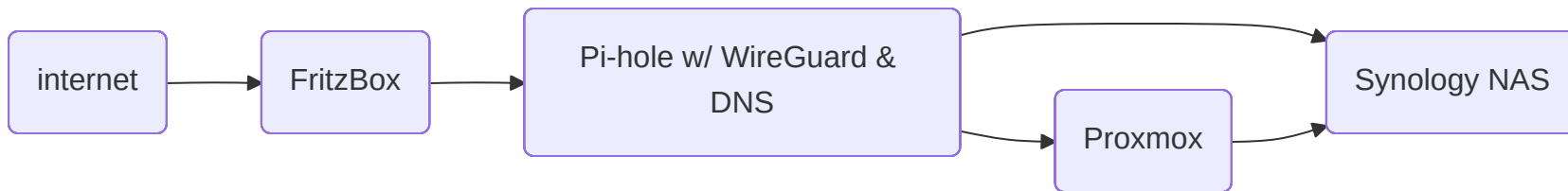
# Domain Hosting



■ I learned something about ``CNAME`` and ``A`` entries



## Home infrastructure (1/x)





## Home infrastructure (2/x)



- But wait, there is a NAS and a Proxmox server?
- Let's talk about hardware





## Hardware (1/2): NAS

NAS: with 4 HDDs

- costs: mainly the HDDs (1000 EUR)
- space: 8TB
- Synology's "RAID-5"





## Hardware (2/2): Server

- costs: 320 EUR
- CPU: Intel Alder Lake N95
- RAM: **32GB** DDR4 (3200MHz)
- SSD: **1TB** M.2 NVMe PCIe 3.0 M.2 2280 (**Max 2TB**)



rufus

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[run Windows?](#)

[Does it have built-in Wi-Fi?](#)



DreamQuest Mini PC 195 Processor 32GB +1TB SSD Mini Desktop PC 4K@60Hz/Gigabit Ethernet Suitable for Daily Office Use

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|                  |            |
|------------------|------------|
| Brand            | DreamQuest |
| Operating system | One Size   |
| CPU model        | Celeron    |
| CPU speed        | 3.4 GHz    |
| Cache size       | 1 TB       |
| Graphics card    | Integrated |



# Access to Homesever

- VPN / Wireguard
- Public access
- Geoblocking



# Why did I choose Proxmox?

Alternatives:

- kubernetes
- nix

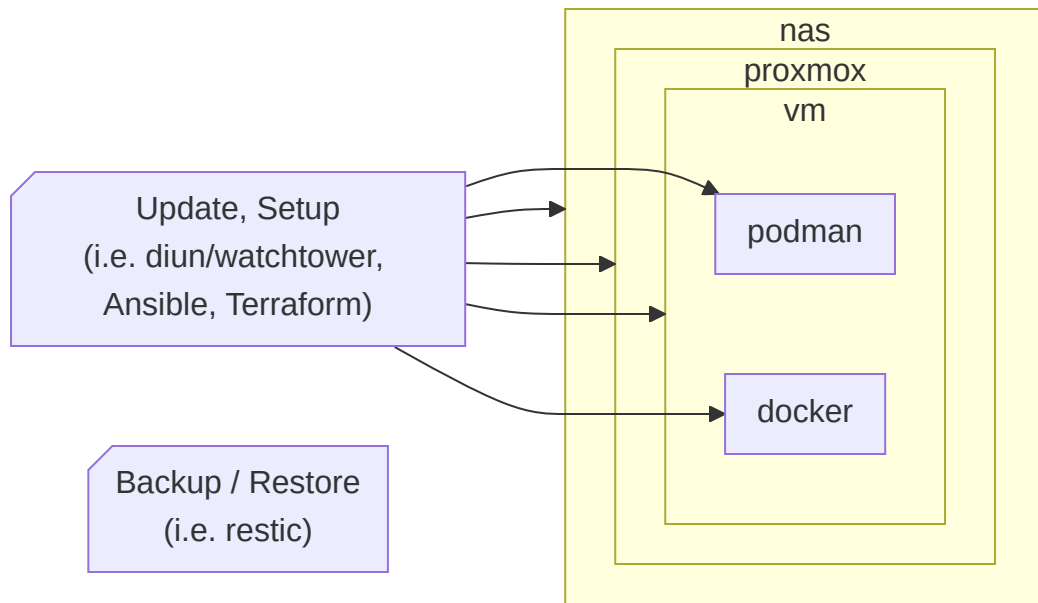
Both are currently too complex for me.

I picked a different rabbit hole:

- Proxmox with VMs and
  - with Ansible
  - with Terraform (OpenTofu)
- Proxmox VMs provide natural containment → safety feature



# Overview





# Discussion



# Thank You!

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-  <https://github.com/draptik>
-  <https://draptik.github.io/talks/>
-  <https://floss.social/@drechsler>
-  <https://bsky.app/profile/drechsler.bsky.social>
-  <https://www.linkedin.com/in/patrick-drechsler-draptik/>

Slides 

- QR Code or
- <https://draptik.github.io/2025-09-seneca-selfhosting/>

